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A NEW CLASS OF ITERATIVE METHODS FOR NONSELFADJOINT OR INDEFINITE PROBLEMS*

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Abstract. A new technique is proposed to solve NSPD (nonsymmetric or indefinite) problems that are "compact" perturbations of some SPD (symmetric positive definite) problems. In the new algorithm, a direct method is first used to solve the original equation restricted on a coarser space (that has a considerably smaller dimension), then an SPD equation for the residue is solved by using one or a few iterations of a given iterative algorithm. It is shown that for any convergent iterative method for the SPD problem, the new algorithm always converges with essentially the same rate if the coarse space is properly chosen. In applications, for multiplicative domain decomposition methods, the algorithm consists of solving the original NSPD problem on the coarse mesh and solving SPD equations on all subdomains; for multigrid methods, except when the correction on the coarsest mesh is first performed for the original NSPD equation, all smoothings are carried out for SPD equations on all other levels. It is shown that most of these algorithms converge uniformly provided that the coarsest meshsize is reasonably small (but independent of the fine meshsize). Discussions are also devoted to some popular algorithms such as SOR method.

Key words. conjugate gradient, domain decomposition, finite element, indefinite, multigrid, nonsymmetric, SOR, space decomposition, subspace correction

AMS(MOS) subject classifications. 65F10, 65N30

1. Introduction. In this paper we shall develop some new iterative methods for algebraic systems arising from the discretization of the nonsymmetric or indefinite partial differential equations (NPSD). A prototype of such equations is

$$\begin{cases} \hat{\mathcal{L}}U = F & \text{in } \Omega, \\ U = 0 & \text{on } \partial\Omega, \end{cases} \tag{1.1}$$

where Ω is a bounded domain in R^d, and $\hat{\mathcal{L}} = \mathcal{L} + \mathcal{N}$ with

$$\mathcal{L}U = -\sum_{i,j=1}^{d} \frac{\partial}{\partial x_j}\left(a_{ij}(x)\frac{\partial U}{\partial x_i}\right) \quad \text{and} \quad \mathcal{N}U = \sum_{i=1}^{d} b_i(x)\frac{\partial U}{\partial x_i} + c(x)U.$$

We assume all the coefficients are sufficiently smooth and the matrix $(a_{ij}(x))$ is symmetric and uniformly positive definite for any $x \in \Omega$. We also assume that (1.1) is uniquely solvable for any $F \in L^2(\Omega)$ (a space consisting of all square integrable functions on Ω).

When finite element or finite difference methods are used to discretize equations like (1.1), the resultant algebraic systems are in general not symmetric positive definite (SPD). Hence the solution of these equations becomes more difficult. It is well known that the solution of an SPD system is often much easier. Algorithms such as SOR, conjugate gradient methods and preconditioning techniques, loosely speaking, mostly work well for SPD systems. More contemporary algorithms such as domain decomposition and multigrid methods are better developed and better understood for SPD systems.

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We note that what makes the equation not SPD are those lower-order terms in the equation. In other words, the leading term that dominates the equation is SPD. It is then a natural attempt to design methods by using the SPD leading term. One simple idea is to consider the normal equation and use the preconditioned conjugate gradient method by means of the preconditioner for the SPD problem corresponding to $\mathcal{L}$. An example of the work along this line is [5]. But such an approach is not always desirable since the condition number of the system is essentially squared.

The approach in this paper is completely different. In our iterative process for solving the discretized system of $\hat{\mathcal{L}}$, we first solve (exactly) the equation restricted on a much smaller subspace (defined on a coarse mesh). The resultant residue, as far as we understand from our experience in multigrid theories [15], mostly contains high frequencies, whereas on these high frequencies the discretized $\hat{\mathcal{L}}$ and $\mathcal{L}$ are very "close." Hence a further correction of this residue can be switched to the discretized SPD equation for $\mathcal{L}$ (to smooth the high frequencies).

The idea in this paper may be traced back to Cai and Widlund [6], where an additive Schwarz domain decomposition method is studied. Such an idea was generalized by Xu and Cai in [16]. Both of these works are only for the construction of preconditioners for GMRES-type algorithms. The theory in this paper is different and the GMRES-type algorithms are not necessary. In some sense, our algorithms are multiplicative variants of those in [16]. We do not particularly aim for the construction of preconditioners, but rather for the design of plain linear iterative methods. Nevertheless, preconditioners can result from our algorithms for the use of GMRES-type algorithms. As a result of our new theory, a whole class of new algorithms can be obtained. Some difficult problems in domain decomposition and multigrid methods can be resolved quite easily by our theory. It seems that the approach in this paper is quite general and a wide range of applications is expected. As an example, we shall discuss applications to second-order elliptic equations.

The remainder of the paper is organized as follows. In §2 we shall present the basic algorithm and its convergence theory. Section 3 is devoted to the study of iterative algorithms based on space decomposition and subspace corrections. In §4 a number of new multigrid and domain decomposition methods will be presented and analyzed; methods like SOR will also be discussed.

2. The basic algorithm and theory. In this section, we shall present the basic algorithm, analyze its convergence properties, and discuss its application with GMRES-type algorithms.

2.1. Preliminaries. Our theory will be established in a given linear vector space $\mathcal{V}$. We assume that $\mathcal{V}$ is equipped with a norm $\|\cdot\|$, which is induced by an inner product $(\cdot,\cdot)$, and $\hat{A}$ is a given invertible linear operator on $\mathcal{V}$. We are interested in solving the equation

$$\hat{A}u = f \tag{2.1}$$

for a given $f \in \mathcal{V}$.

We shall assume that $\hat{A}$ is a perturbation of a linear SPD operator. More specifically

$$\hat{A} = A + N.$$

Here the operator $A : \mathcal{V} \mapsto \mathcal{V}$ is SPD in the sense that

$$(Au, v) = (u, Av) \quad \forall u, v \in \mathcal{V} \quad \text{and} \quad (Av, v) > 0 \quad \text{if } v \neq 0;$$

the perturbation operator $N : \mathcal{V} \mapsto \mathcal{V}$ is not SPD in general. $(A\cdot,\cdot)$ defines an inner product on $\mathcal{V}$ and induces a norm on $\mathcal{V}$, denoted by $\|\cdot\|_A$. Given an operator $G : \mathcal{V} \mapsto \mathcal{V}$, we define its A-norm by

$$\|G\|_A = \sup_{v\in\mathcal{V}} \frac{\|Gv\|_A}{\|v\|_A}.$$

We shall develop a class of iterative algorithms for (2.1) in the following form.

ALGORITHM A.

$$u^{k+1} = u^k + \hat{B}(f - \hat{A}u^k)$$

for $k = 0, 1, 2, \cdots$, and any initial guess u^0.

The key element of the above algorithm is the operator $\hat{B} : \mathcal{V} \mapsto \mathcal{V}$. We shall call this operator to be an *iterator* of $\hat{A}$. The iterator $\hat{B}$ may be thought of as an approximation of $\hat{A}^{-1}$ or a preconditioner of $\hat{A}$, and it is not necessarily linear. The construction of an iterative method like Algorithm A then amounts to the construction of the operator $\hat{B}$. We note that a sufficient condition for the convergence of Algorithm A is

$$\rho = \|I - \hat{B}\hat{A}\|_A < 1.$$

In this case

$$\|u - u^k\|_A \le \rho^k \|u\|_A.$$

We shall call the number ρ a *convergence factor.*

One of the main ingredients for Algorithm A will be a subspace $\mathcal{V}_0 \subset \mathcal{V}$. Assuming $\mathcal{V}_0 \subset \mathcal{V}$ is given, we define a linear operator $\hat{A}_0 : \mathcal{V}_0 \mapsto \mathcal{V}_0$, and two projections $Q_0, \hat{P}_0 : \mathcal{V} \mapsto \mathcal{V}_0$ by

$$\begin{aligned} &(\hat{A}_0 u_0, v_0) = (\hat{A}u_0, v_0) \quad \forall u_0, v_0 \in \mathcal{V}_0, \\ &(\hat{A}\hat{P}_0 u, v_0) = (\hat{A}u, v_0), \qquad (Q_0 u, v_0) = (u, v_0) \quad \forall u \in \mathcal{V}, v_0 \in \mathcal{V}_0, \end{aligned}$$

respectively. It is clear that $\hat{A}_0$ and Q_0 are well defined. We shall assume that $\hat{A}_0$ is invertible, which implies that $\hat{P}_0$ is also well defined.

By the definitions of $\hat{P}_0, \hat{A}_0$, and Q_0,

$$\hat{A}_0\hat{P}_0 = Q_0\hat{A}. \tag{2.2}$$

It follows that, for a given $f \in \mathcal{V}$,

$$\hat{u}_0 = \hat{A}_0^{-1} Q_0 f \quad \text{if and only if } (\hat{A}\hat{u}_0, v_0) = (f, v_0) \quad \forall v_0 \in \mathcal{V}_0.$$

Another main ingredient of our theory is the SPD system of the form

$$Av = g. \tag{2.3}$$

The new algorithm to be presented is built upon a given iterative method for the above SPD equation. In other words, we assume that an iterator of A, denoted by B, is given. For convenience, the corresponding algorithm will be referred to as Algorithm S, as follows.

ALGORITHM S.

$$v^{i+1} = v^i + B(g - Av^i)$$

for $i = 0, 1, 2, \cdots$, and for any initial guess v^0.

2.2. Iterative algorithms and convergence analysis. By means of a given subspace $\mathcal{V}_0$ and an iterator B of A (namely, the given Algorithm S), the main algorithm of this paper is defined as follows.

ALGORITHM N. Given $u^0 \in \mathcal{V}$, assume u^k is defined for $k \geq 0$. Then

1. Solve (exactly) the equation on $\mathcal{V}_0$:

$$\hat{A}_0 \hat{u}_0 = Q_0(f - \hat{A}u^k). \tag{2.4}$$

2 Solve the SPD equation (2.3) with $g = f - \hat{A}(u^k + \hat{u}_0)$ by Algorithm S for $i = 0, 1, \cdots, p$, and zero initial guesses to get v^p.

3 $u^{k+1} = u^k + \hat{u}_0 + v^p$.

In light of the terminologies in multigrid methods, the first step of the above algorithm is the correction on the coarse space $\mathcal{V}_0$; the second step plays the role of smoothing by the SPD operator A.

Remark 2.1. In applications, $\dim \mathcal{V}_0 \ll \dim \mathcal{V}$. Hence $\hat{u}_0$ in the first step of the above algorithm may be solved by a direct method. It seems desirable to get and store an LU factorization of $\hat{A}_0$ since (2.4) is repeated in each iteration. Roundoff error in this process is not a problem, namely, $\hat{u}_0$ does not have to be very accurate. Theoretically, for example, if $\tilde{u}_0$ is the computed value of $\hat{u}_0$ and $\epsilon = \|\hat{u}_0 - \tilde{u}_0\|_A$, then the convergence factor (see the estimate (2.9) below) of the whole algorithm is at most affected by a constant multiple of ϵ, which is negligible.

Let us now derive the error equation of the above algorithm. For simplicity, we first take $p = 1$. Note that $f = \hat{A}u$. It follows that

$$\hat{u}_0 = \hat{P}_0(u - u^k) \quad \text{and} \quad v^1 = B\hat{A}(I - \hat{P}_0)(u - u^k).$$

Thus

$$u - u^{k+1} = (I - B\hat{A})(I - \hat{P}_0)(u - u^k). \tag{2.5}$$

For general p,

$$u - u^{k+1} = (I - (I - (I - BA)^p)A^{-1}\hat{A})(I - \hat{P}_0)(u - u^k).$$

Obviously, the Algorithm N is identical to Algorithm A if the iterator $\hat{B}$ satisfies

$$I - \hat{B}\hat{A} = (I - (I - (I - BA)^p)A^{-1}\hat{A})(I - \hat{P}_0). \tag{2.6}$$

The convergence theory of Algorithm N will be established in terms of the following parameter:

$$\delta_0 = \sup_{u,v \in \mathcal{V}} \frac{(N(I - \hat{P}_0)u, v)}{\|u\|_A \|v\|_A}. \tag{2.7}$$

The relationships among the space $\mathcal{V}$, subspace $\mathcal{V}_0$, and operators $\hat{A}$ and A will all be characterized by this parameter. The assumption that we shall make is that δ_0 can be sufficiently small by choosing the subspace $\mathcal{V}_0$ properly.

Note that, for any $u, v \in \mathcal{V}$,

$$(N(I - \hat{P}_0)u, v) \leq \delta_0 \|u\|_A \|v\|_A, \qquad (N(I - \hat{P}_0)u, v) \leq \delta_0 \|(I - \hat{P}_0)u\|_A \|v\|_A. \tag{2.8}$$

Here the first inequality follows from the definition and the second follows from the first with $(I - \hat{P}_0)u$ in place of u.

We now present the basic estimate of the paper.

THEOREM 2.1. *Assume* $\delta_0 < 1$; *then*

$$\|\hat{E}\|_A \le \sigma. \tag{2.9}$$

Here $\hat{E} = I - \hat{B}\hat{A}$, *with* $\hat{B}$ *given by* (2.6), *is the error operator of Algorithm* N *and*

$$\sigma = \rho^p + \rho_0, \quad \rho = \|I - BA\|_A, \quad \rho_0 = \frac{4\delta_0}{1-\delta_0}. \tag{2.10}$$

Note that $\rho_0 \to 0$ as $\delta_0 \to 0$. The proof of the above theorem will be given a little later. We shall first state its consequence.

THEOREM 2.2. *Under the assumption of Theorem* 2.1,

$$\|u - u^k\|_A \le (\rho^p + \rho_0)^k \|u - u^0\|_A,$$

where u^k *are defined by Algorithm* N *and* u *is the solution of* (2.1). *Therefore Algorithm* N *is convergent if* $\rho < 1$ *and* δ_0 *is sufficiently small so that* $\rho_0 < 1 - \rho^p$.

Remark 2.2. The assumption $\rho < 1$ is a sufficient condition for the convergence of Algorithm S. This condition is in fact necessary sometimes (for example, when B is SPD). Therefore Theorem 2.2 essentially means that Algorithm N for NSPD problems is always convergent as long as Algorithm S for the SPD equation is convergent.

The rest of this section is devoted to the proof of Theorem 2.1. First we need a simple lemma that gives an estimate of $I - \hat{P}_0$.

LEMMA 2.3. *Assume* $\delta_0 < 1$. *Then*

$$\|u - \hat{P}_0 u\|_A \le (1 + \eta_0)\|u\|_A \quad \forall u \in \mathcal{V},$$

where

$$\eta_0 = \frac{2\delta_0}{1-\delta_0}.$$

Proof. Denote $u_0 = \hat{P}_0 u$. By definition and (2.8)

$$\begin{aligned}
\|u - u_0\|_A^2 &= (A(u-u_0), u-u_0) = (\hat{A}(u-u_0), u) - (N(u-u_0), u-u_0) \\
&= (A(u-u_0), u) + (N(u-u_0), u) - (N(u-u_0), u-u_0) \\
&\le \|u-u_0\|_A \|u\|_A + \delta_0 \|u-u_0\|_A (\|u\|_A + \|u-u_0\|_A) \\
&\le (1+\delta_0)\|u-u_0\|_A\|u\|_A + \delta_0\|u-u_0\|_A^2.
\end{aligned}$$

It follows that, if $\delta_0 < 1$,

$$\|u - u_0\|_A \le \frac{1+\delta_0}{1-\delta_0}\|u\|_A$$

as desired. □

We are now in a position to give a proof of Theorem 2.1.

Proof. Without loss of generality, we assume that $p = 1$. Given $u \in \mathcal{V}$, denote $u_0 = \hat{P}_0 u, v = A^{-1}\hat{A}(u - u_0)$, and $w = u - u_0$. We shall first show that

$$\|w - v\|_A \le \delta_0 \|u\|_A, \quad \|v\|_A \le (1 + \delta_0 + \eta_0)\|u\|_A. \tag{2.11}$$

In fact,

$$\begin{aligned}\|w-v\|_A^2 &= (A(w-v), w-v) = ((A-\hat{A})(u-u_0), w-v)\\ &= -(N(u-u_0), w-v)\\ &\le \delta_0\|u\|_A\|w-v\|_A.\end{aligned}$$

The first estimate in (2.11) then follows. To see the second estimate in (2.11), by Lemma 2.3,

$$\begin{aligned}\|v\|_A^2 &= (Av, v) = (\hat{A}(u-u_0), v)\\ &= (A(u-u_0), v) + (N(u-u_0), v)\\ &\le (1+\eta_0)\|u\|_A\|v\|_A + \delta_0\|u\|_A\|v\|_A\\ &\le (1+\delta_0+\eta_0)\|u\|_A\|v\|_A.\end{aligned}$$

Therefore (2.11) is justified.

Thanks to (2.11), the rest of the proof is easy:

$$\begin{aligned}\|(I-B\hat{A})(I-\hat{P}_0)u\|_A &= \|w-B(Av)\|_A\\ &\le \|w-v\|_A + \|v-B(Av)\|_A \le \delta_0\|u\|_A + \rho\|v\|_A\\ &\le (\delta_0+\rho(1+\delta_0+\eta_0))\,\|u\|_A \le (\rho+2\delta_0+\eta_0)\,\|u\|_A\end{aligned}$$

as desired. □

2.3. Preconditioners for GMRES-type methods. In this section we shall demonstrate how our algorithms may be used in conjunction with GMRES-type methods (c.f. [9], [12]). The idea is to use iterator $\hat{B}$ as a preconditioner.

Unlike the conjugate gradient method for SPD problem, which is always convergent, the GMRES method may not be convergent without proper preconditioning. Hence a preconditioner for the GMRES method is not only to speed up the convergence but more importantly to guarantee the convergence as well.

Recall that, for σ given by (2.10),

$$\|I-\hat{B}\hat{A}\|_A \le \sigma,$$

which implies that

$$A((I-\hat{B}\hat{A})v, v) \le \|(I-\hat{B}\hat{A})v\|_A\|v\|_A \le \sigma A(v,v).$$

Thus

$$(1-\sigma)A(v,v) \le A(\hat{B}\hat{A}v, v).$$

On the other hand, it is trivial that

$$\|\hat{B}\hat{A}v\|_A \le (1+\sigma)\|v\|_A.$$

Using the convergence theory in [9], we conclude from the above two inequalities that if the GMRES method is applied to the preconditioned system

$$\hat{B}\hat{A}u = \hat{B}f$$

with the inner product $A(\cdot,\cdot)$, the algorithm is convergent with a convergence factor

$$1-\frac{(1-\sigma)^2}{(1+\sigma)^2}$$

provided that $\sigma < 1$.

Notice that

$$\hat{B}\hat{A} = \hat{P}_0 + B\hat{A}(I-\hat{P}_0). \tag{2.12}$$

In [16] we have constructed a preconditioner $\tilde{B}$ that satisfies

$$\tilde{B}\hat{A} = \hat{P}_0 + \beta B\hat{A}. \tag{2.13}$$

Here $\beta > 0$ is a positive parameter. The preconditioner $\hat{B}$ can be regarded as the multiplicative version of $\tilde{B}$.

The explicit expressions of $\hat{B}$ and $\tilde{B}$ are given as follows:

$$\hat{B} = (I-B\hat{A})\hat{A}_0^{-1}Q_0 + B, \qquad \tilde{B} = \hat{A}_0^{-1}Q_0 + \beta B.$$

2.4. Algorithms in terms of vectors and matrices. In this section the algorithms studied earlier will be presented in terms of vectors and matrices, which are more convenient to code.

Assume that $\{\phi_i\}_{i=1}^n$ is a given basis of $\mathcal{V}$ and $\{\phi_i^0\}_{i=1}^{n_0}$ is a given basis of $\mathcal{V}_0$. The stiffness matrices of operator $\hat{A}$ and A on these bases are given by

$$\hat{\mathcal{A}} = ((\hat{A}\phi_i,\phi_j)) \in \mathbb{R}^{n\times n}, \quad \hat{\mathcal{A}}_0 = ((\hat{A}\phi_i^0,\phi_j^0)) \in \mathbb{R}^{n_0\times n_0}, \quad A = ((A\phi_i,\phi_j)) \in \mathbb{R}^{n\times n}.$$

The equation (2.1) is then equivalent to

$$\hat{\mathcal{A}}x = b. \tag{2.14}$$

The relation between (2.1) and (2.14) is given by

$$u = \sum_{i=1}^n x_i\phi_i, \qquad b_i = (f,\phi_i).$$

Since $\mathcal{V}_0 \subset \mathcal{V}$, there are $t_{ij} \in \mathbb{R}$ such that

$$\phi_j^0 = \sum_{i=1}^n t_{ij}\phi_i, \qquad 1 \le j \le n_0.$$

In this way, we get a matrix $T = (t_{ij}) \in \mathbb{R}^{n\times n_0}$, which agrees with the prolongation matrix, and its transpose, T^t, agrees with the restriction matrix used in multigrid methods.

Using the notation introduced above, Algorithm N can be written in the following equivalent form.[1]

Given $x^0 \in \mathbb{R}^n$, assume x^k is defined for $k \ge 0$. Then

1. Solve (exactly) the equation of order n_0 :

$$\hat{\mathcal{A}}_0\hat{x}_0 = T^t(b-\hat{\mathcal{A}}x^k).$$

2. Solve the SPD equation $\mathcal{A}z = b - \hat{\mathcal{A}}(x^k + T\hat{x}_0)$ by a given iterative method for p times (corresponding to Algorithm S) to get z^p with zero initial guess.
3. $x^{k+1} = x^k + T\hat{x}_0 + z^p$.

[1] We refer to [14] for some simple techniques to see their equivalence.

3. Space decomposition and subspace correction. In this section we shall first discuss Algorithm S for the SPD problem and then the corresponding Algorithm N and its modification.

According to [14], many linear iterative methods for SPD problems can be formulated by space decomposition and subspace corrections. We shall investigate how this type of algorithm can fit into our theory.

3.1. Algorithms for SPD problems. In this section we shall review the algorithms for SPD problems based on space decomposition and subspace correction.

All the methods to be discussed are built upon a decomposition of $\mathcal{V}$ that consists of a number of subspaces $\mathcal{V}_i \subset \mathcal{V}$, $i = 0, \cdots, J$, such that

$$\mathcal{V} = \sum_{i=0}^{J} \mathcal{V}_i. \tag{3.1}$$

This means that, for each $v \in \mathcal{V}$, there exist $v_i \in \mathcal{V}_i, i = 0, 1, 2, \cdots, J$, such that $v = \sum_{i=0}^{J} v_i$. This representation of v may not be unique in general.

For each i, Q_i and P_i will denote the orthogonal projections from $\mathcal{V}$ to $\mathcal{V}_i$ with respect to the inner products $(\cdot,\cdot)$ and $(\cdot,\cdot)_A$, respectively, such that

$$(Q_i u, v_i) = (u, v_i), \quad (P_i u, v_i)_A = (u, v_i)_A, \quad u \in \mathcal{V}, v_i \in \mathcal{V}_i, \tag{3.2}$$

and $A_i : \mathcal{V}_i \mapsto \mathcal{V}_i$ is an SPD operator defined by

$$(A_i u_i, v_i) = (A u_i, v_i), \qquad u_i, v_i \in \mathcal{V}_i.$$

An analogy of (2.2) is

$$A_i P_i = Q_i A. \tag{3.3}$$

The algorithm for (2.3) will be designed based on given algorithms for solving the subspace equations $A_i u_i = f_i$ for $f_i \in \mathcal{V}_i$. Again these algorithms are characterized by some iterators of A_i, denoted by R_i.

ALGORITHM M. Given $v^0 \in \mathcal{V}$, assume that $v^k \in \mathcal{V}$ is obtained. Then v^{k+1} is defined by

$$v^{k+(i+1)/(J+1)} = v^{k+i/(J+1)} + R_i Q_i (f - A v^{k+i/(J+1)})$$

for $i = 0, 1, \cdots, J$.

Denote $T_i = R_i Q_i A$. By (3.3), $T_i = R_i A_i P_i$. A direct manipulation gives that

$$v - v^{k+1} = (I - BA)(v - v^k),$$

where

$$I - BA = (I - T_J)(I - T_{J-1}) \cdots (I - T_1)(I - T_0). \tag{3.4}$$

Again we remark that with the definition of B by the above identity, Algorithms S and M are equivalent.

Among various convergence estimates available in the literature for Algorithm M, we quote a simple qualitative result.

LEMMA 3.1. *Assume that, for each i, R_i is* SPD *and*

$$\sigma(R_i A_i) \subset (0, 2).$$

Then Algorithm M *is convergent and*

$$\rho = \|I - BA\|_A < 1.$$

The proof of the above theorem can be found in [14] and is implicitly contained in [4]. The estimate of ρ depends on the subspaces $\mathcal{V}_i$ and their interactions, subspace solvers R_i, etc. The efficiency of the algorithm depends on how far ρ bounds away from 1. For detailed quantitative estimates for ρ, we refer to [4] and [14].

3.2. Algorithm N and its modifications. With the subspace correction methods given above, Algorithm N can then be defined accordingly. By Theorems 2.1 and 2.2 and Lemma 3.1, this algorithm is convergent if $\sigma(R_iA_i) \subset (0,2)$ for each i and δ_0 is sufficiently small. In the rest of this section we shall use the special structures of these methods to make certain modifications for Algorithm N. The point here is to save a little bit of work in the algorithm.

Suppose that $\mathcal{V}_0$ used in the definition of Algorithm N coincides with that in the decomposition (3.1). Then, if Algorithm N is applied directly with Algorithm M, the subspace problems on $\mathcal{V}_0$ are solved twice in each iteration, once for A_0 and once for $\hat{A}_0$. A natural question is whether one of them can be removed from the algorithm without affecting its convergence property. We shall provide an affirmative answer to this question. It is actually quite clear that only the one for A_0 can be removed since the role played by $\hat{A}_0$ is absolutely crucial.

With the removal of the solver for A_0 from Algorithm M, we have the modified Algorithm N as follows.

ALGORITHM ND. Given $u^0 \in \mathcal{V}$, assume that u^k is defined for $k \geq 1$. Then we define $u^{k+1} = \hat{u}^k + v^J$ where

$$\hat{u}^k = u^k + \hat{A}_0^{-1}Q_0(f - \hat{A}u^k)$$

and, for $i = 1, \cdots, J$,

$$v^i = v^{i-1} + R_iQ_i(g - Av^{i-1})$$

with $g = f - \hat{A}\hat{u}^k$ and $v^0 = 0$.

The error equation of the above algorithm is

$$u - u^{k+1} = (I - \tilde{B}\hat{A})(I - \hat{P}_0)(u - u^k)$$

where

$$I - \tilde{B}A = (I - T_J)(I - T_{J-1})\cdots(I - T_1).$$

We now give the convergence estimate for Algorithm ND.

THEOREM 3.2. *Assume that $\sigma(R_iA_i) \subset (0,2)$ for each $1 \leq i \leq J$. Then Algorithm* ND *converges if δ_0, given by* (2.7), *is sufficiently small. Furthermore, for the error operator of the algorithm $\tilde{E} = (I - \tilde{B}\hat{A})(I - \hat{P}_0)$,*

$$\|\tilde{E}\|_A \leq \rho + \frac{6\delta_0}{1 - \delta_0}.$$

Proof. Obviously, for B defined by (3.4) with $R_0 = A_0^{-1}$,

$$I - BA = (I - \tilde{B}A)(I - P_0). \tag{3.5}$$

A direct manipulation yields

$$(I-\tilde{B}\hat{A})(I-\hat{P}_0)=(I-B\hat{A})(I-\hat{P}_0)+(I-\tilde{B}A)(P_0-\hat{P}_0)+(B-\tilde{B})N(I-\hat{P}_0).$$

Thus

$$\begin{aligned}\|(I-\tilde{B}\hat{A})(I-\hat{P}_0)u\|_A &\le \|(I-B\hat{A})(I-\hat{P}_0)u\|_A \\ &\quad +\|(I-\tilde{B}A)(P_0-\hat{P}_0)u\|_A+\|(B-\tilde{B})N(I-\hat{P}_0)u\|_A \\ &\equiv I_1+I_2+I_3.\end{aligned}$$

The estimate of I_1 is given by Theorem 2.1:

$$I_1\le(\rho+\rho_0)\|u\|_A.$$

It follows from the definitions of $\hat{P}_0$ and P_0 that

$$(A(\hat{P}_0-P_0)u,v_0)=(N(I-\hat{P}_0)u,v_0)\quad \forall u\in\mathcal{V},v_0\in\mathcal{V}_0,$$

which, with $v_0=(\hat{P}_0-P_0)u$, implies

$$\|(\hat{P}_0-P_0)u\|_A\le\delta_0\|u\|_A. \tag{3.6}$$

By the assumption on R_i, $\|I-T_i\|_A\le 1$, which implies that $\|I-\tilde{B}A\|_A\le 1$. Hence

$$I_2\le\|(P_0-\hat{P}_0)u\|_A\le\delta_0\|u\|_A.$$

It remains to estimate I_3. We first note that, by (3.5),

$$(B-\tilde{B})A=(I-\tilde{B}A)P_0.$$

Thus

$$\|(B-\tilde{B})A\|_A=\|I-\tilde{B}A\|_A\|P_0\|_A\le 1.$$

Let "t" and "$*$" denote the transpositions with respect to the inner products $(\cdot,\cdot)$ and $(A\cdot,\cdot)$, respectively; then

$$\|(B-\tilde{B})^tA\|_A=\|[(B-\tilde{B})A]^*\|_A=\|(B-\tilde{B})A\|_A\le 1.$$

Consequently,

$$\begin{aligned}\|(B-\tilde{B})N(I-\hat{P}_0)u\|_A^2 &= ((B-\tilde{B})N(I-\hat{P}_0)u,A(B-\tilde{B})N(I-\hat{P}_0)u) \\ &\le\delta_0\|u\|_A\|(B-\tilde{B})^tA(B-\tilde{B})N(I-\hat{P}_0)u\|_A \\ &\le\delta_0\|u\|_A\|(B-\tilde{B})^tA\|_A\|(B-\tilde{B})N(I-\hat{P}_0)u\|_A \\ &\le\delta_0\|u\|_A\|(B-\tilde{B})N(I-\hat{P}_0)u\|_A.\end{aligned}$$

Hence

$$I_3=\|(B-\tilde{B})N(I-\hat{P}_0)u\|_A\le\delta_0\|u\|_A.$$

The desired estimate then follows. □

4. Second-order elliptic equations. In this section we shall discuss some applications of our algorithms to second-order elliptic boundary value problems (1.1). Domain decomposition and multigrid methods will be the main examples.

To consider the finite element discretization for (1.1), we need first to transfer the equation to an equivalent variational formulation. To this end, let $H^1(\Omega)$ be the standard Sobolev space consisting of square integrable functions with square integrable (weak) derivatives of first order and $H_0^1(\Omega)$ a subspace of $H^1(\Omega)$ consisting of functions that vanish on $\partial\Omega$ (in an appropriate sense). Then $U \in H_0^1(\Omega)$ is the solution of (1.1) if and only if

$$\hat{A}(U,\chi) = (F,\chi) \quad \forall \chi \in H_0^1(\Omega), \tag{4.1}$$

where

$$\hat{A}(U,\chi) = \int_\Omega \sum_{i,j=1}^d a_{ij}(x)\frac{\partial U}{\partial x_i}\frac{\partial \chi}{\partial x_j} + \sum_{i=1}^d b_i(x)\frac{\partial U}{\partial x_i}\chi + c(x)U\chi \quad \text{and} \quad (F,\chi) = \int_\Omega F\chi.$$

We first define the finite element approximation scheme. Assume that Ω is a polyhedral domain and that Ω has been triangulated with $\Omega = \cup_i \tau_i$, where τ_i's are simplices of size h with $h \in (0,1]$ and quasiuniform. By this we mean that there exist constants C_0 and C_1 not depending on h such that each simplex τ_i is contained in (contains) a ball of radius $C_1 h$ (respectively, $C_0 h$). The finite element space $\mathcal{V}_h$ is defined to be the functions that are continuous on Ω, piecewise linear with respect to the triangulation $\{\tau_i\}$, and vanish on $\partial\Omega$.

In the sequel, we shall drop the subscript h and denote $\mathcal{V} = \mathcal{V}_h$.

The finite element approximation to the solution of (4.1) is the function $u \in \mathcal{V}$ satisfying

$$\hat{A}(u,v) = (F,v) \quad \forall v \in \mathcal{V}. \tag{4.2}$$

This equation is uniquely solvable if h is sufficiently small (cf. [13]).

Define a linear operator $\hat{A} : \mathcal{V} \mapsto \mathcal{V}$ by

$$(\hat{A}u, v) = \hat{A}(u,v), \qquad u,v \in \mathcal{V}. \tag{4.3}$$

The equation (4.2) is then equivalent to

$$\hat{A}u = f \tag{4.4}$$

with some $f \in \mathcal{V}$.

If we define

$$A(U,\chi) = \int_\Omega \sum_{i,j=1}^d a_{ij}(x)\frac{\partial U}{\partial x_i}\frac{\partial \chi}{\partial x_j},$$

an SPD operator $A : \mathcal{V} \mapsto \mathcal{V}$ can be defined similarly. Introducing the Galerkin projection $\hat{P}_h : H_0^1(\Omega) \mapsto \mathcal{V}$ defined by

$$\hat{A}(\hat{P}_h v, \phi) = \hat{A}(v,\phi) \quad \forall v \in H_0^1(\Omega), \quad \phi \in \mathcal{V}.$$

Then $u = \hat{P}_h U$, where U and u are solutions of (4.2) and (4.2), respectively.

Following Schatz [13], there is a constant $\alpha \in (0,1]$ (depending on the regularity of the adjoint equation of (1.1)) such that

$$\|v - \hat{P}_h v\|_{H^{1-\alpha}(\Omega)} \leq Ch^{\alpha}\|v\|_{H^1(\Omega)} \quad \forall v \in H_0^1(\Omega). \tag{4.5}$$

Here $H^{1-\alpha}(\Omega)$ is the standard fractional Sobolev space for $\alpha < 1$ and $H^0(\Omega) = L^2(\Omega)$.

We shall take the space $\mathcal{V}_0 = \mathcal{V}_{h_0}$, the finite element space defined on a triangulation with meshsize $h_0 \in (h, 1)$, which satisfies

$$\mathcal{V}_{h_0} \subset \mathcal{V}_h.$$

Correspondingly, $\hat{P}_0 = \hat{P}_{h_0}$. We assume that h_0 is sufficiently small so that $\hat{A}_0$ is invertible and hence $\hat{P}_0$ is well defined. Applying (4.5) with h_0 in place of h, we then get by integration by parts

$$(N(I - \hat{P}_0)u, v) \leq C\|(I - \hat{P}_0)u\|\,\|v\|_A \leq c_0 h_0^{\alpha}\|u\|_A\|v\|_A. \tag{4.6}$$

Hence for δ_0 defined by (2.7),

$$\delta_0 \leq c_0 h_0^{\alpha}. \tag{4.7}$$

Remark 4.3. If other boundary conditions such as Neumann conditions are imposed on the partial differential equation, the estimate of δ_0 is a little different. Instead of (4.6), we have, for $\alpha < \frac{1}{2}$,

$$(N(I - \hat{P}_0)u, v) \leq C\|(I - \hat{P}_0)u\|_{H^{1-\alpha}(\Omega)}\|v\|_A \leq c_0 h_0^{\alpha}\|u\|_A\|v\|_A.$$

The constraint that $\alpha < \frac{1}{2}$ is due to the trace inequality used in the derivation of the above inequality. In this case, (4.7) holds for $\alpha < \frac{1}{2}$.

4.1. Multiplicative domain decomposition methods. In this section, we shall apply our theory to study the multiplicative domain decomposition methods for (1.1). This type of method has been well studied for SPD problems (cf. [4]), but little is known for (1.1). The algorithm to be proposed is not a direct generalization of those for SPD problems. In our algorithm, all subdomain problems are solved for SPD equations, and the original NSPD equation is solved only on the coarse mesh.

We start by assuming that we are given a set of overlapping subdomains $\{\Omega_i\}_{i=1}^J$ of Ω whose boundaries align with the mesh triangulation defining $\mathcal{V}$. One way of defining the subdomains and the associated partition is by starting with disjoint open sets $\{\Omega_i^0\}_{i=1}^J$, with $\bar{\Omega} = \cup_{i=1}^J \bar{\Omega}_i^0$ and $\{\Omega_i^0\}_{i=1}^J$ quasiuniform of size h_0. The subdomain Ω_i is defined to be a mesh subdomain containing Ω_i^0, with the distance from $\partial\Omega_i \cap \Omega$ to Ω_i^0 greater than or equal to ch_0 for some prescribed constant c.

Based on the subdomains given above, we can then define subspace $\mathcal{V}_i$ for $i = 1, 2, \cdots, J$ by

$$\mathcal{V}_i = \{v \in \mathcal{V}, v(x) = 0 \text{ if } x \notin \Omega_i\}.$$

In addition, we introduce a coarse finite element subspace $\mathcal{V}_0 \subset \mathcal{V}$ defined from a quasi-uniform triangulation of Ω of size h_0. We then consider the following space decomposition

$$\mathcal{V} = \sum_{i=0}^{J} \mathcal{V}_i. \tag{4.8}$$

With the space decomposition given, Algorithm M in §3 can then be defined accordingly. The convergence of this algorithm has been analyzed in [4].

LEMMA 1 [4]. *Algorithm* M *associated with the decomposition* (4.8) *satisfies*

$$\|E_J\|_A^2 \le 1 - \frac{\omega_0(2-\omega_1)}{C}.$$

Here $\omega_0 \in (0,1)$ *and* $\omega_1 \in (0,2)$ *are constants*[2] *such that* $\sigma(R_iA_i) \subset (\omega_0, \omega_1)$ *for all* i, *and* C *is a constant independent of the number of subdomains* J *and the meshsize* h.

Combining the above lemma with Theorem 3.2, we get Theorem 4.2.

THEOREM 4.1. *Under the assumptions in* Lemma 4.1, *Algorithm* N *associated with the decomposition* (4.8) *satisfies*

$$\|E_J\|_A^2 \le 1 - \frac{\omega_0(2-\omega_1)}{C} + ch_0^\alpha, \tag{4.9}$$

where c *and* C *are positive constants independent of the number of subdomains* J *and the meshsize* h. *Consequently, Algorithm* N *converges uniformly if* $h_0 < \epsilon_0$ *for some constant* $\epsilon_0 \in (0,1)$ *independent of* h *and* J.

Algorithm M associated with the decomposition (4.8) may be regarded as a "multiplicative" version of the "additive" preconditioner (for GMRES method) proposed by Cai and Widlund [6].

Remark 4.4. If we apply Algorithm M directly without removing the solver for A_0, the corresponding Algorithm N satisfies

$$\|\tilde{E}_J\|_A^2 \le \left(1 - \frac{\omega_0(2-\omega_1)}{C}\right)^p + ch_0^\alpha.$$

We can increase the value of p to relax the constraint on h_0. It pays to use this algorithm instead of Algorithm ND if we do not want h_0 to be too small.

A note on a more straightforward multiplicative algorithm. It is natural to consider a direct generalization of Algorithm M for the NSPD equation (4.4) (with $\hat{A}$ and $\hat{R}_i \approx \hat{A}_i^{-1}$ in place of A and R_i). This algorithm does not fit into the framework of this paper, but it can be analyzed by the techniques developed in [4]. We shall now consider a simple case that subdomain and coarse space problems are solved exactly.

ALGORITHM ND0. Given $u^0 \in \mathcal{V}$, assume that u^k is defined for $k \ge 1$. Then we define $u^{k+1} = v^J$ where, for $i = 0, 1, \cdots, J$,

$$v^i = v^{i-1} + \hat{A}_i^{-1} Q_i (f - \hat{A} v^{i-1})$$

with $v^0 = u^k$.

The error operator of the above algorithm is given by

$$\hat{E}_J = (I - \hat{P}_J)(I - \hat{P}_{J-1}) \cdots (I - \hat{P}_1)(I - \hat{P}_0). \tag{4.10}$$

We shall now use a simple perturbation argument to establish a simple result. For a more general and refined analysis, we refer to Cai and Widlund [7].

PROPOSITION 4.2. *Algorithm* ND0 *is uniformly convergent if* h_0 *and* h^α/h_0 *are sufficiently small.*

Proof. We first note that combining (3.6) with (4.7) gives that

$$\|\hat{P}_0 - P_0\|_A \le Ch_0^\alpha.$$

[2] Note that $\omega_0 = \omega_1 = 1$ if $R_i = A_i^{-1}$ for each i.

Similarly, by a simple scaling argument, we can show that

$$\|\hat{P}_k - P_k\|_A \le Ch_0 h^\alpha, \qquad 1 \le k \le J.$$

Here $\hat{P}_k : \mathcal{V} \mapsto \mathcal{V}_k$ is defined by

$$(\hat{A}\hat{P}_k u, v_k) = (\hat{A}u, v_k) \quad \forall u \in \mathcal{V}, v_k \in \mathcal{V}_k.$$

Consequently, we can easily see that

$$\|\hat{E}_k\|_A \le \prod_{i=0}^{k}(1 + \|\hat{P}_i - P_i\|_A) \le 1 + Ckh_0 h^\alpha \le 1 + C\frac{h^\alpha}{h_0}$$

since $k \le J = O(h_0^{-2})$.

Denote

$$\hat{E}_k = (I - \hat{P}_k)(I - \hat{P}_{k-1}) \cdots (I - \hat{P}_1)(I - \hat{P}_0),$$

$$E_k = (I - P_k)(I - P_{k-1}) \cdots (I - P_1)(I - P_0).$$

It is straightforward to see that

$$\hat{E}_k - E_k = (I - P_k)(\hat{E}_{k-1} - E_{k-1}) + (P_k - \hat{P}_k)\hat{E}_{k-1}.$$

Therefore

$$\begin{aligned}\|\hat{E}_k - E_k\|_A &\le \|\hat{E}_{k-1} - E_{k-1}\|_A + \|P_k - \hat{P}_k\|_A \|\hat{E}_{k-1}\|_A \\ &\le \|\hat{E}_{k-1} - E_{k-1}\|_A + Ch_0 h^\alpha \left(1 + \frac{h^\alpha}{h_0}\right).\end{aligned}$$

Consequently,

$$\begin{aligned}\|\hat{E}_J - E_J\|_A &\le \|\hat{P}_0 - P_0\|_A + \sum_{k=1}^{J} Ch_0 h^\alpha \left(1 + \frac{h^\alpha}{h_0}\right) \\ &\le C\left(h_0^\alpha + \frac{h^\alpha}{h_0}\left(1 + \frac{h^\alpha}{h_0}\right)\right).\end{aligned}$$

By Lemma 4.1, $\|E_J\|_A$ is uniformly bounded by 1; hence $\|\hat{E}_J\|_A$ is uniformly bounded by 1 if h_0 and h^α/h_0 are sufficiently small. This completes the proof. □

4.2. Multigrid algorithm. As an application of the theory in the previous section, new multigrid algorithms will now be proposed for (1.1). Roughly speaking, in our algorithm, the original NSPD equation is solved only on the coarsest mesh, whereas the SPD equations are smoothed on all other levels.

As is done earlier, we approximate the problem (4.1) by finite element discretizations on a quasi-uniform triangulation $\mathcal{T}$ of Ω. We assume the triangulation $\mathcal{T}$ is constructed by a successive refinement process. More precisely, $\mathcal{T} = \mathcal{T}_J$ for some $J > 1$, and $\mathcal{T}_k$ for $k \le J$, are a nested sequence of quasi-uniform triangulations that consist of simplices $\mathcal{T}_k = \{\tau_k^i\}$ of size h_k for $k = 1, \cdots, J$, such that $\Omega = \cup_i \tau_k^i$, where the quasi-uniformity constants are independent of k (cf. [8]). These triangulations should be nested in the sense that any simplex τ_{k-1}^l can be written as a union of

simplices of $\{\tau_k^i\}$. We further assume that there is a constant $\eta > 1$, independent of k, such that h_k is proportional to η^{-k}. As an example, in the two-dimensional case, a finer grid is obtained by connecting the midpoints of the edges of the triangles of the coarser grid, with $\mathcal{T}_0$ being the given coarsest initial quasi-uniform triangulation.

Corresponding to each triangulation $\mathcal{T}_k$, a finite element space $\mathcal{V}_k$ can be defined by $\mathcal{V}_k = \{v \in H_0^1(\Omega) : v|_\tau \in \mathcal{P}_1(\tau) \text{ for all } \tau \in \mathcal{T}_k\}$, where $\mathcal{P}_1$ is the space of linear polynomials. Obviously,

$$\mathcal{V}_0 \subset \mathcal{V}_1 \subset \cdots \subset \mathcal{V}_J \equiv \mathcal{V}. \tag{4.11}$$

Using the usual multigrid methods for SPD problems [15], we then get the corresponding Algorithm N in an obvious way. Corresponding to Algorithm ND, we have the following algorithm.

ALGORITHM NM. Given u^0, for $l = 0, 1, \cdots$,

$$\begin{aligned} u^{l+\frac{1}{2}} &= u^l + \hat{A}_0^{-1} Q_0 (f - \hat{A} u^l), \\ u^{l+1} &= u^{l+\frac{1}{2}} + B_J (f - \hat{A} u^{l+\frac{1}{2}}), \end{aligned}$$

where B_J is defined by the usual multigrid method for SPD problems on the multilevel spaces in (4.11) with the removal of the correction on $\mathcal{V}_0$.

Let us now give an example algorithm to define the multigrid operator B_J (by induction with respect to J).

ALGORITHM DEFINING B_J. $B_1 = R_1$. Assume that $B_{k-1} : \mathcal{V}_{k-1} \mapsto \mathcal{V}_{k-1}$ has been defined and then $B_k : \mathcal{V}_k \mapsto \mathcal{V}_k$ for $g \in \mathcal{V}_k$ is defined by

$$B_k g = v^{m_k},$$

where

1. For $w^0 = 0$ and $l = 1, 2, \cdots, p$,

$$w^l = w^{l-1} + B_{k-1} Q_{k-1} (g - A_{k-1} w^{l-1});$$

2. For $v^0 = w^p$ and $i = 1, 2, \cdots, m_k$,

$$v^i = v^{i-1} + R_k (g - A_k v^{i-1}).$$

Per convention, the algorithm is called V-cycle if $p = 1$ and W-cycle if $p = 2$. In most cases $m_k = m$ for all k. An exception is that m_k increases as k decreases, for example, $m_k = 2^{J-k} m$ and $p = 1$, which is known as variable V-cycle (cf. [2]).

Following [3], Algorithm NM can be formulated in the form of Algorithm ND. The next theorem is derived from Theorem 3.2 by using the multigrid convergence results for SPD problems (cf. [3], [15], and [14]).

THEOREM 4.3. *Assume the smoothers R_k, for $1 \le k \le J$, are given, for example, by the Gauss–Seidel method. For V-cycle, variable V-cycle, or W-cycle, there are constants $c > 0$ and $\rho \in (0, 1)$ that are independent of mesh parameters such that*

$$\|I - \hat{B}\hat{A}\|_A \le \rho + c h_0^\alpha.$$

Consequently, these algorithms are all uniformly convergent if h_0 is sufficiently small.

It is known that multigrid methods can be applied directly to (1.1). Theories in [1], [11], [15], and [5] have shown that these multigrid algorithms converge provided that h_0 is sufficiently small. It would be interesting to compare them numerically

with our new algorithm. As far as the theoretical analysis is concerned, our algorithm is much simpler, and furthermore, some difficult problems can be resolved with ease. As an example, using the theories in [4] for multigrid algorithms to SPD problems with local mesh refinement, corresponding algorithms and theories to nonsymmetric or indefinite problems can be set up in an obvious way provided that the coarsest mesh is quasiuniform.

4.3. Other solvers. As we mentioned before, the underlying SPD problem in our algorithms can be solved by any iterative method. A typical example is the preconditioned conjugate gradient method. In this respect, any preconditioning techniques can be used, such as additive Schwarz methods and hierarchical basis preconditioners. In such cases, the operator B may not be linear, but our theory still applies. If we prefer to use some simpler but less effective methods for the solution of the SPD problem, we may need a sufficiently large p.

We shall now, as an illustration, discuss the SOR method (cf. [10]) for the Poisson equation on unit square. In this case, if the optimal relaxation parameter is used,

$$\rho = \|I - BA\|_A \leq 1 - c_1 h.$$

Here c_1 is some positive constant. Therefore the convergence factor for the corresponding Algorithm N, which will be referred to below as NSOR, satisfies

$$\delta \leq (1 - c_1 h)^p + c_0 h_0^\alpha.$$

This means that the algorithm converges only if p is sufficiently large and h_0 is sufficiently small.

We shall now roughly compare NSOR with SOR for their convergence behavior and computing complexity. For this purpose, we take an integer p_h and $\epsilon_0 > 0$ such that

$$(1 - c_1 h)^{p_h} \leq \frac{1}{2}, \qquad c_0 \epsilon_0^\alpha \leq \frac{1}{10}.$$

Then, after mp_hth iteration, for SOR we have

$$\|v - v^{mp_h}\|_A \leq (1 - c_1 h)^{mp_h} \|v\|_A \leq 0.5^m \|v\|_A,$$

and for NSOR (mth outer iteration with p_hth inner iteration each time), if $h_0 \leq \epsilon_0$, we have

$$\|u - u^m\|_A \leq ((1 - c_1 h)^{p_h} + c_0 h_0^\alpha)^m \|u\|_A \leq 0.6^m \|u\|_A.$$

This seems to indicate that the convergence behaviors of SOR and NSOR are not much different. Compared with SOR, in each iteration, the NSOR method needs to solve one more equation for $\hat{A}_0$, but the amount of work for doing this is very small once an LU factorization of $\hat{A}_0$ is formed (since $\dim \mathcal{V}_0 \ll \dim \mathcal{V}$ for small h). Consequently, the difference in the computational complexity of these two algorithms is insignificant.

Similar arguments also apply when preconditioned conjugate gradient methods are used for the SPD problem with preconditioners such as incomplete Cholesky factorization.

5. Concluding remarks. In conclusion, we have shown that combining a coarse mesh solver with any convergent iterative method for the related SPD problem always gives rise to a convergent algorithm (namely, Algorithm N or ND) for the original NSPD problem. Furthermore, the convergence behavior and computational complexity of the resultant algorithm are quite comparable with those of the corresponding algorithm for the SPD problem. Codes for solving NSPD problems can be easily obtained by simply adding a direct coarse grid solver to any existing codes for SPD problems. Numerical experiments for algorithms such as SOR, multigrid, domain decomposition, and preconditioned conjugate gradient methods will be reported in a future work.

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